

Question 19

1 mark

Identify the trigonometric function that is equivalent to $\sin \frac{\pi}{4} \cos \frac{\pi}{3} + \cos \frac{\pi}{4} \sin \frac{\pi}{3}$.

a) $\sin \frac{2\pi}{7}$

b) $\sin \frac{7\pi}{12}$

c) $\cos \frac{2\pi}{7}$

d) $\cos \frac{7\pi}{12}$

Question 20

1 mark

Identify the logarithmic form of $5^x = 6$.

a) $\log_5 x = 6$

b) $\log_5 6 = x$

c) $\log_6 x = 5$

d) $\log_6 5 = x$

Question 21

1 mark

Given $f(\theta) = 3 \cos 2\theta - 1$ and $g(\theta) = \sin \theta + 1$, identify which statement is true.

- a) Both functions have the same period.
- b) Both functions have the same amplitude.
- c) Both functions have the same minimum value.
- d) Both functions have the same maximum value.

Question 22

1 mark

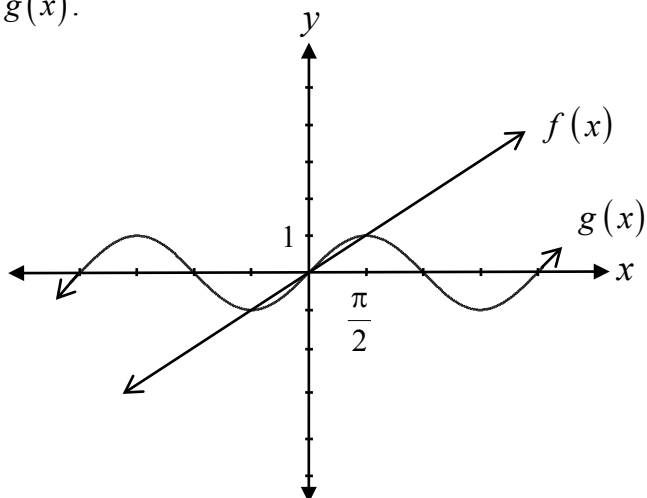
Identify the fourth term in the expansion of $(x + y)^5$.

- a) $10x^4y$
- b) $10x^3y^2$
- c) $10x^2y^3$
- d) $10xy^4$

Question 23

1 mark

Given the graphs of $f(x)$ and $g(x)$, identify the choice with all of the solutions of the equation $f(x) = g(x)$.



a) $x = -2\pi, -\pi, 0, \pi, 2\pi$

b) $x = -\frac{\pi}{2}, 0, \frac{\pi}{2}$

c) $x = \frac{\pi}{2}$

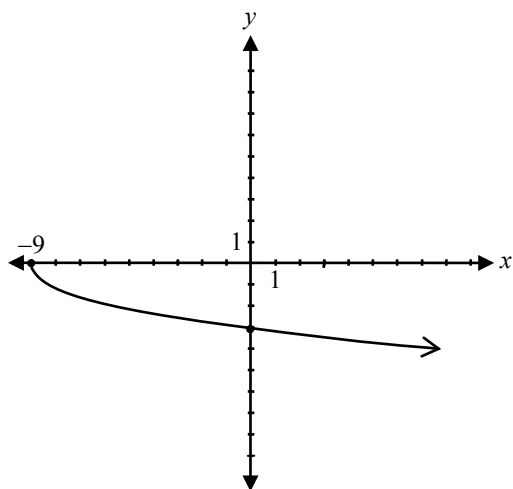
d) $x = -1, 0, 1$

Question 24

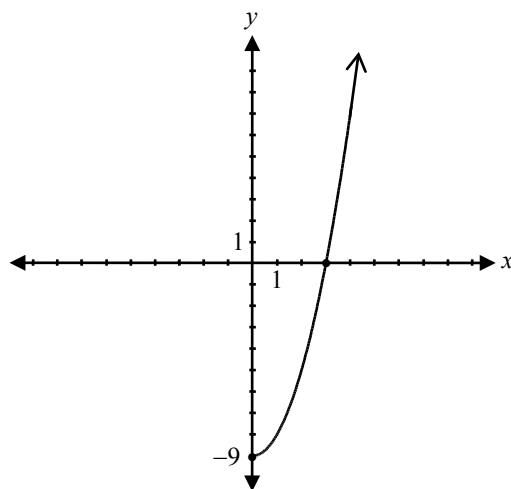
1 mark

Identify the graph of $f^{-1}(x)$ if $f(x) = x^2 - 9, x \geq 0$.

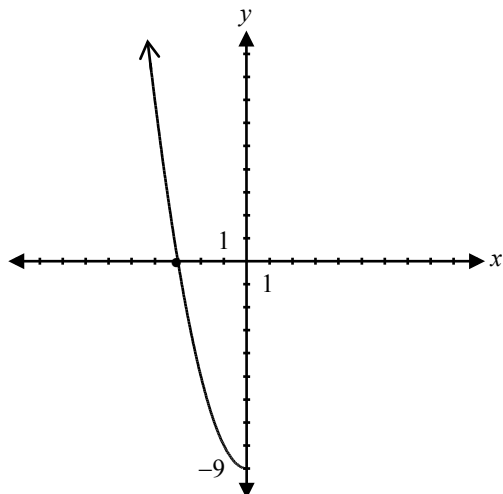
a)



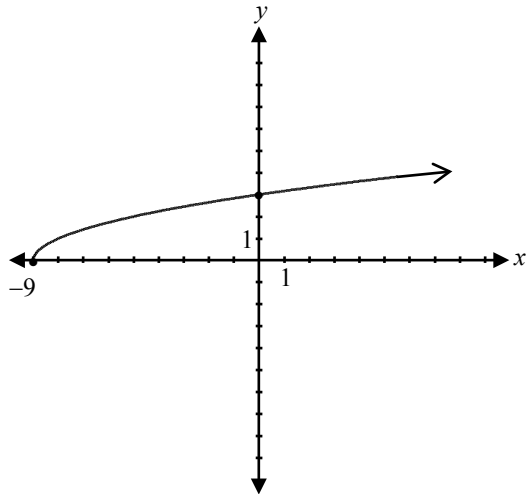
b)



c)



d)



Question 25

1 mark

Using the remainder theorem, identify which value of x results in a remainder of zero given

$$p(x) = x^3 + 7x^2 + 14x + 8.$$

- a) 1
- b) 0
- c) -1
- d) -3

Question 26

1 mark

Evaluate $\cos\left(\cos\left(\frac{3\pi}{2}\right)\right)$.

- a) 1
- b) $\frac{1}{2}$
- c) 0
- d) -1

Question 27

2 marks 121

Match the following equations with their graphs:

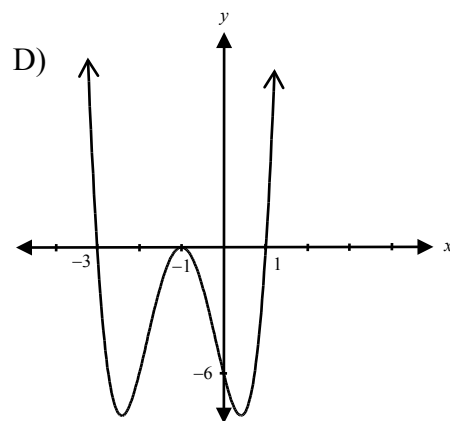
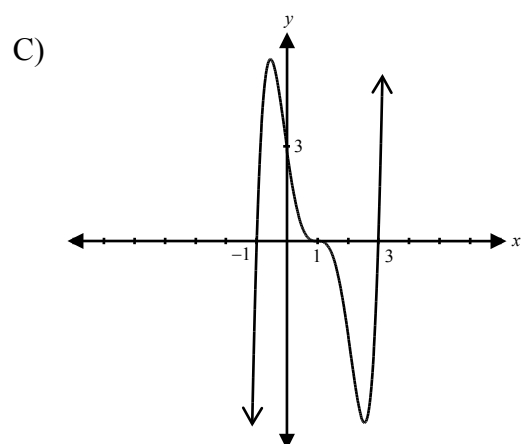
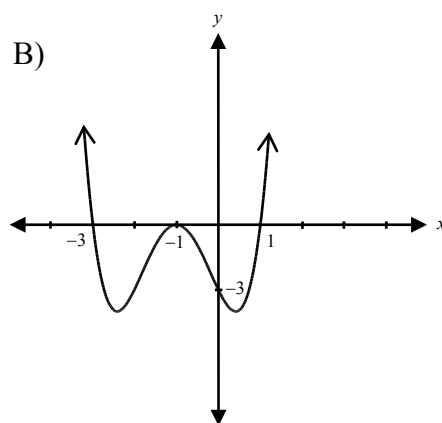
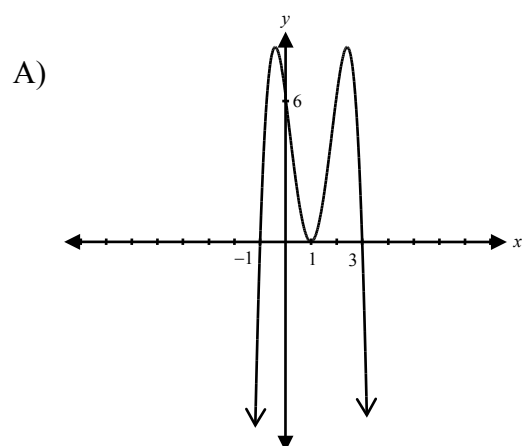
Place the appropriate letter
in this column.

$$f(x) = (x-1)^3(x+1)(x-3) \quad \underline{\hspace{2cm}}$$

$$g(x) = (x+1)^2(x-1)(x+3) \quad \underline{\hspace{2cm}}$$

$$h(x) = -2(x-1)^2(x+1)(x-3) \quad \underline{\hspace{2cm}}$$

$$k(x) = 2(x+1)^2(x-1)(x+3) \quad \underline{\hspace{2cm}}$$



Question 28

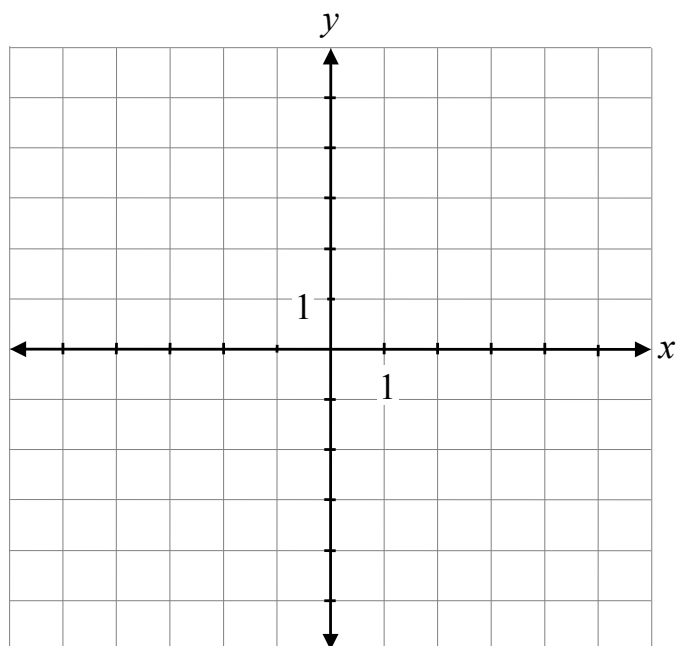
2 marks 122

If $\log 6 = p$, $\log 5 = r$ and $\log 2 = q$, express $\log 60$ in terms of p , q and r .

Question 29

4 marks 123

Sketch the graph of $y = \sqrt{-2x} + 1$.



Question 30

1 mark 124

Justify why the letters of the word FRANCE have a greater number of possible arrangements than the letters of the word CANADA.

Question 31

a) 1 mark b) 3 marks

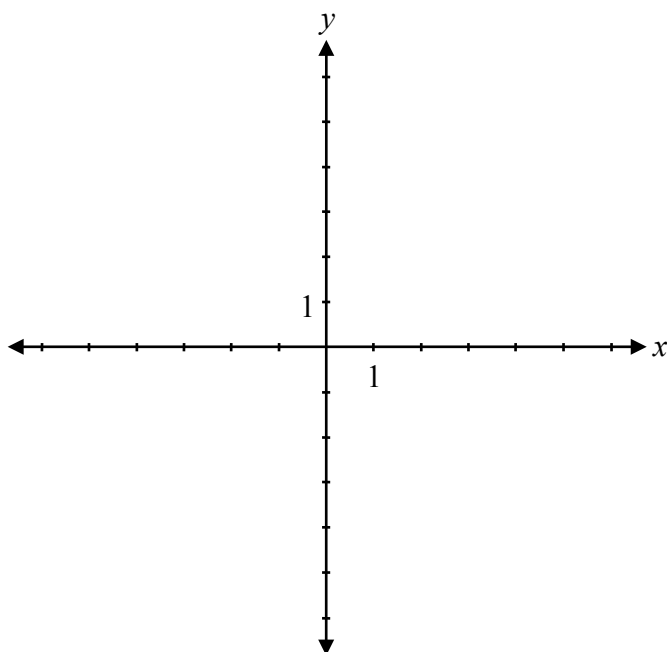
125
126

Given $f(x) = \frac{1}{x-2}$ and $g(x) = x+5$,

a) determine the equation for $f(g(x))$.

$f(g(x)) =$ _____

b) sketch the graph of $f(g(x))$.



Question 32

a) 3 marks b) 1 mark

127
128

Given that $\sin \alpha = \frac{3}{7}$, where α is in Quadrant II, and $\cos \beta = \frac{4}{5}$, where β is in Quadrant IV, determine the exact value of:

a) $\sin(\alpha - \beta)$

b) $\cos 2\alpha$

Question 33

2 marks 129

Determine the domain and range of $f(x) = \sqrt{x-5} - 1$.

Domain: _____

Range: _____

Question 34

1 mark 130

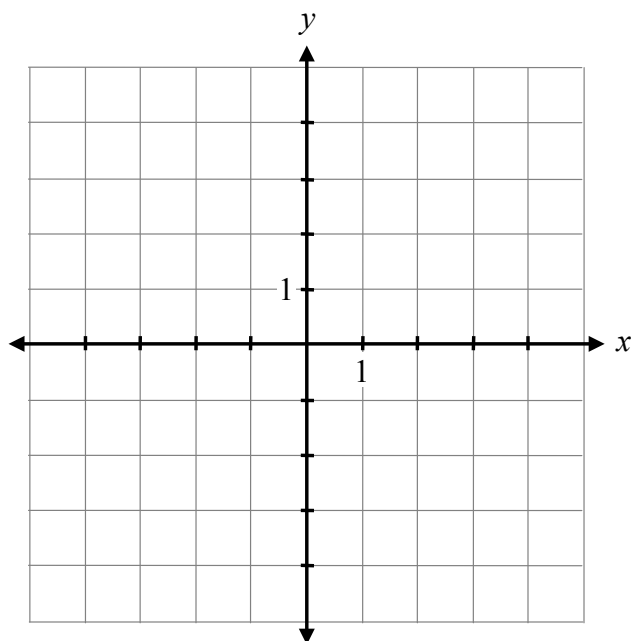
Justify why 4.7 is a better estimate than 4.3 for the value of $\log_2 26$.

Question 35

2 marks 131

Sketch the graph of the function:

$$f(x) = \frac{x(x-2)(x-4)}{(x-2)}$$



Question 36

3 marks 132

Evaluate:

$$\sec^2\left(\frac{\pi}{6}\right) + \tan\left(\frac{7\pi}{6}\right) \csc\left(-\frac{2\pi}{3}\right)$$

Question 37

1 mark 133

The graph of $f(x) = 3x + 7$ is reflected over the y -axis.

Determine the equation of the new function.

$y =$ _____

Question 38

2 marks 134

Determine the equations of all of the asymptotes of the function:

$$y = \frac{2x + 1}{x - 3}$$

Question 39

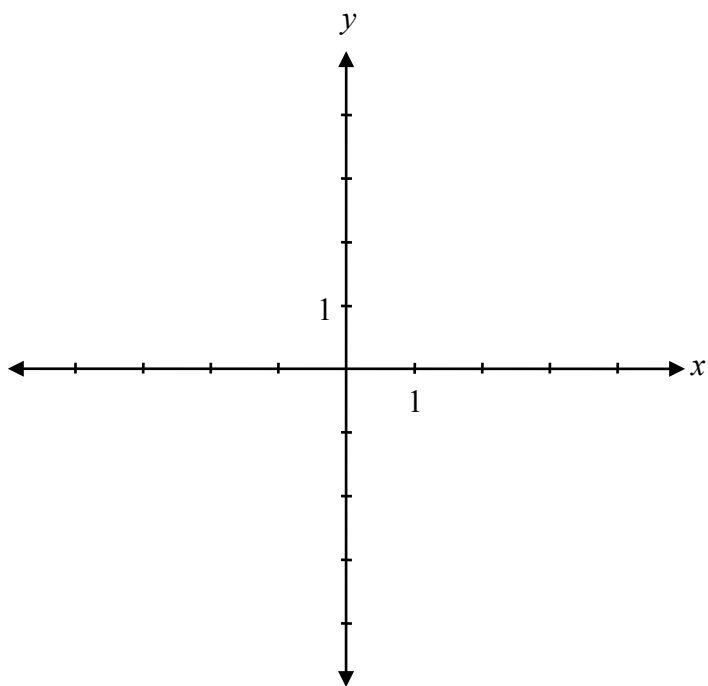
2 marks 135

One of the zeros of $p(x) = x^3 + 6x^2 - 32$ is $x = 2$. Determine all of the other zeros of $p(x)$.

Question 40

3 marks 136

Sketch the graph of $y = -2^x + 2$.



Question 41

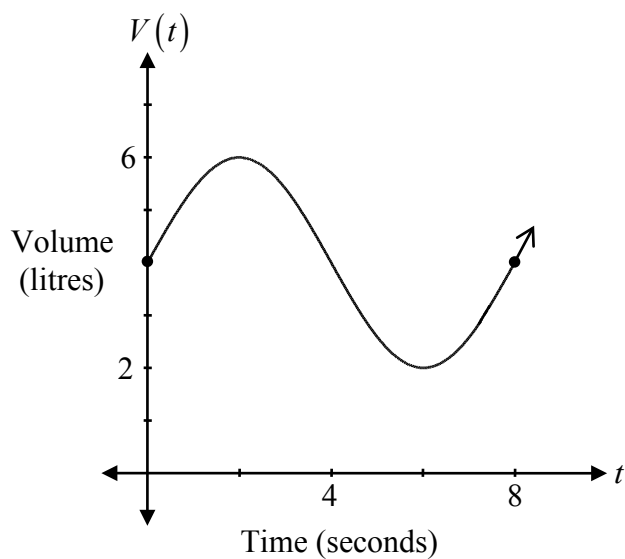
1 mark 137

Given the function $f(x) = \frac{2}{x} - 1$, justify why $f(f(2))$ is undefined.

Question 42

3 marks 138

The following graph represents the volume of air in an adult's lungs. If $V(t)$ is the volume of air in litres and t is the time in seconds, determine an equation that represents this sinusoidal function.



$V(t) =$ _____

Question 43

1 mark 139

Explain why the domain of $y = \log_2(x - 1)$ is $x > 1$.

Question 44

2 marks 140

The point $(-2, 7)$ is on the terminal arm of an angle in standard position.

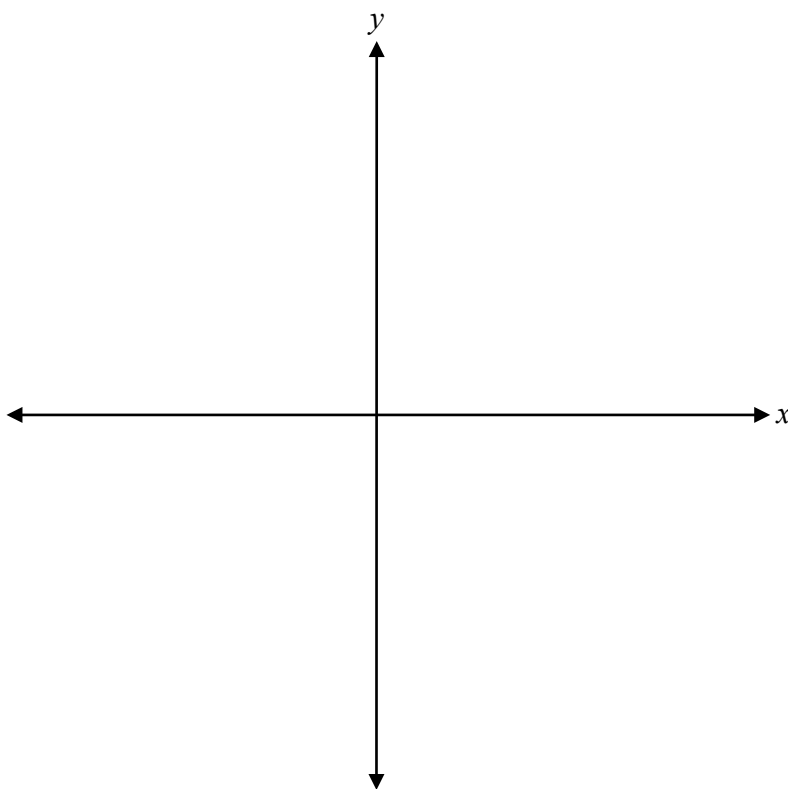
Determine the coordinates of the corresponding point, $P(\theta)$, on the unit circle.

Question 45

3 marks 141

Sketch a graph of $P(x)$ that satisfies all of the following conditions:

- $P(x)$ is a polynomial function of degree 3.
- $P(x)$ has a zero at -3 with a multiplicity of 2.
- $P(x)$ has a zero at 1.
- $P(x)$ has a leading coefficient of -3 .

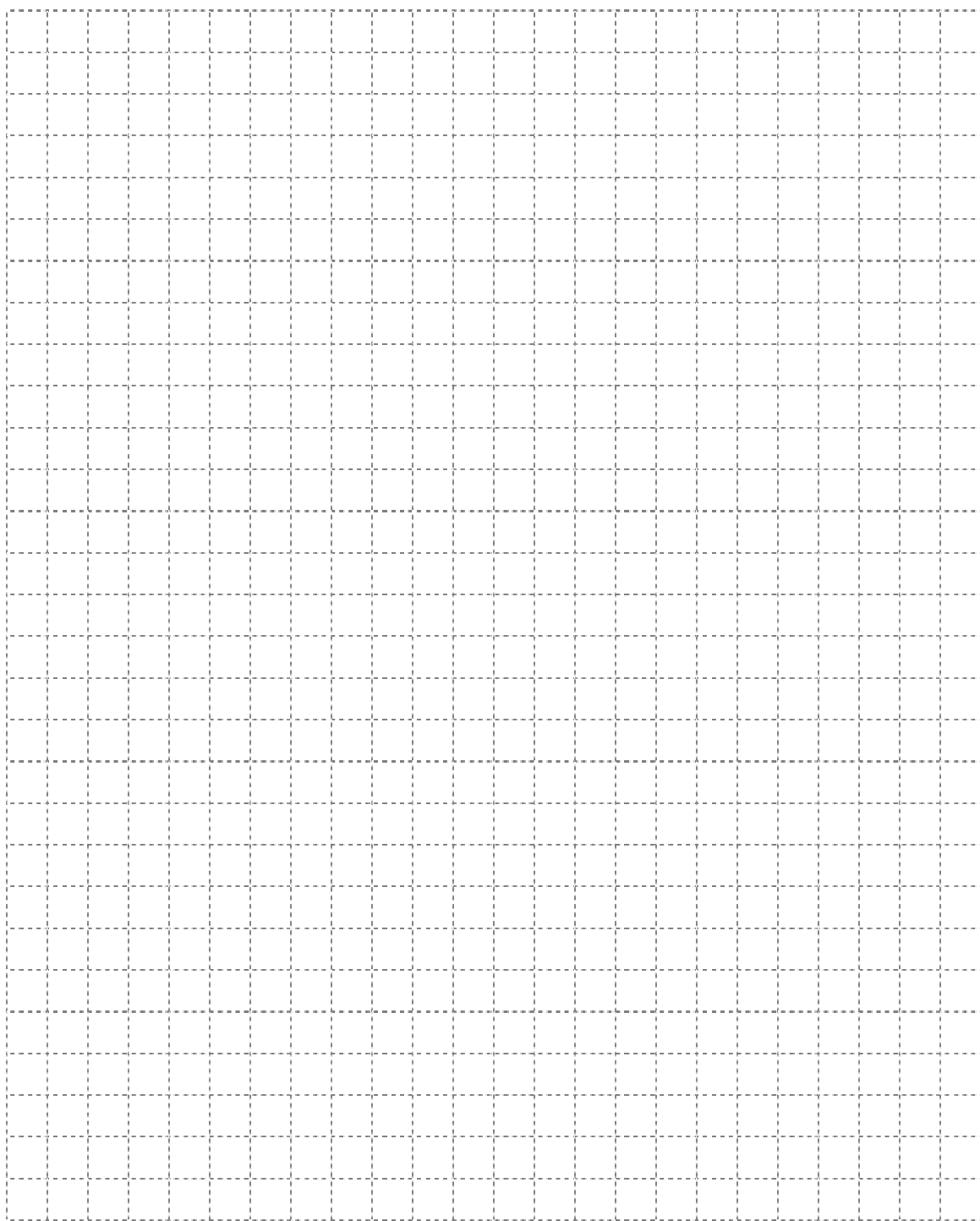


Question 46

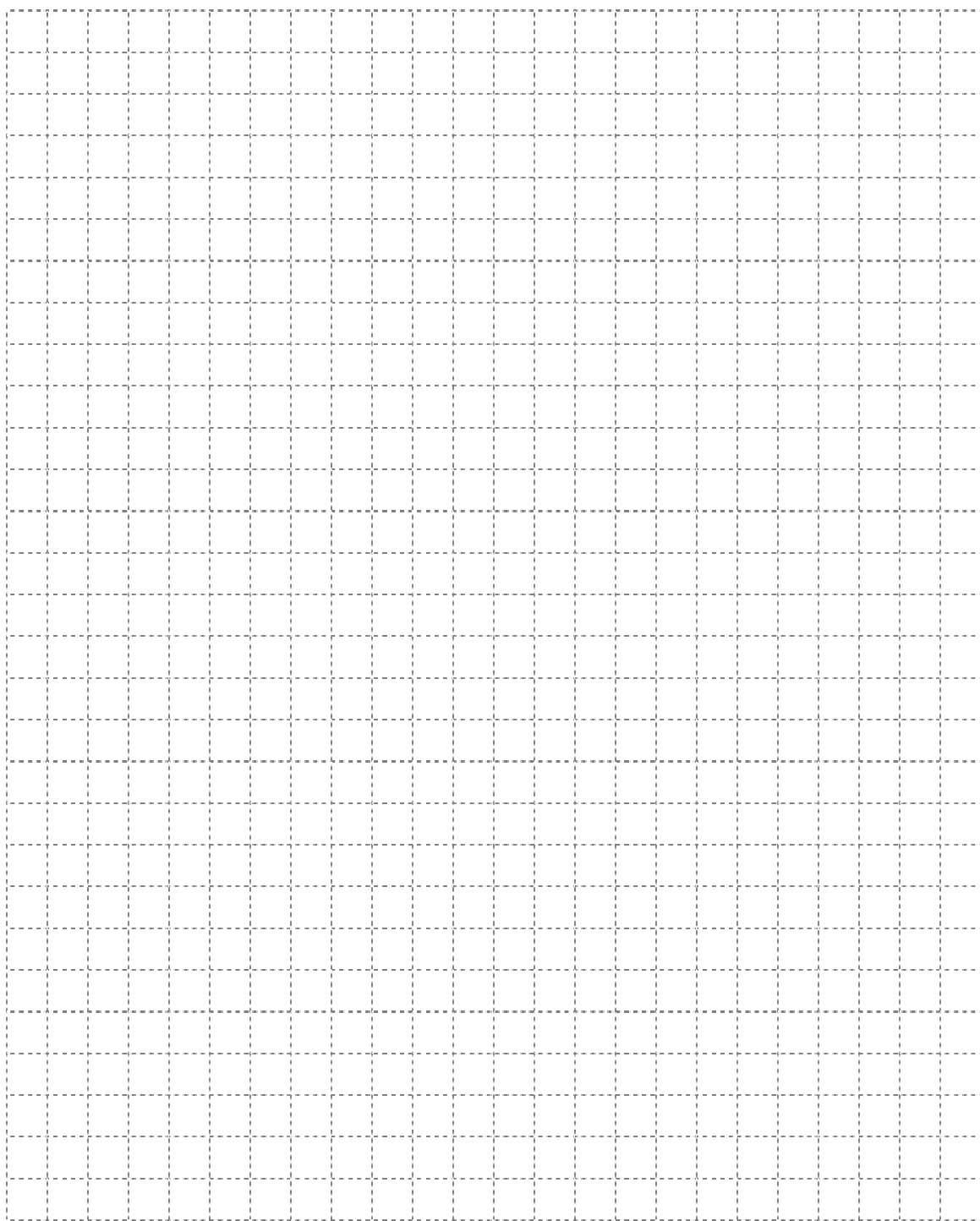
2 marks 142

Determine how many 4-digit numbers greater than 4000 can be made using the digits 2, 3, 4, 5, and 6 if repetitions are not allowed.

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